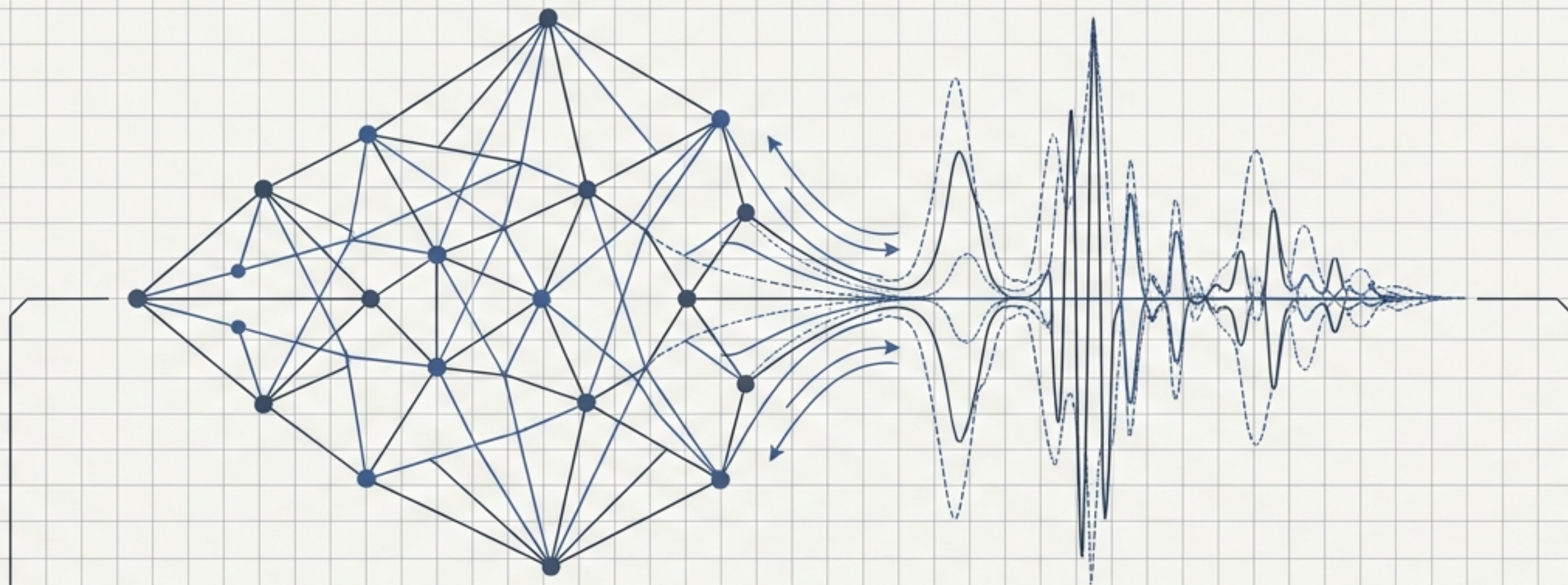




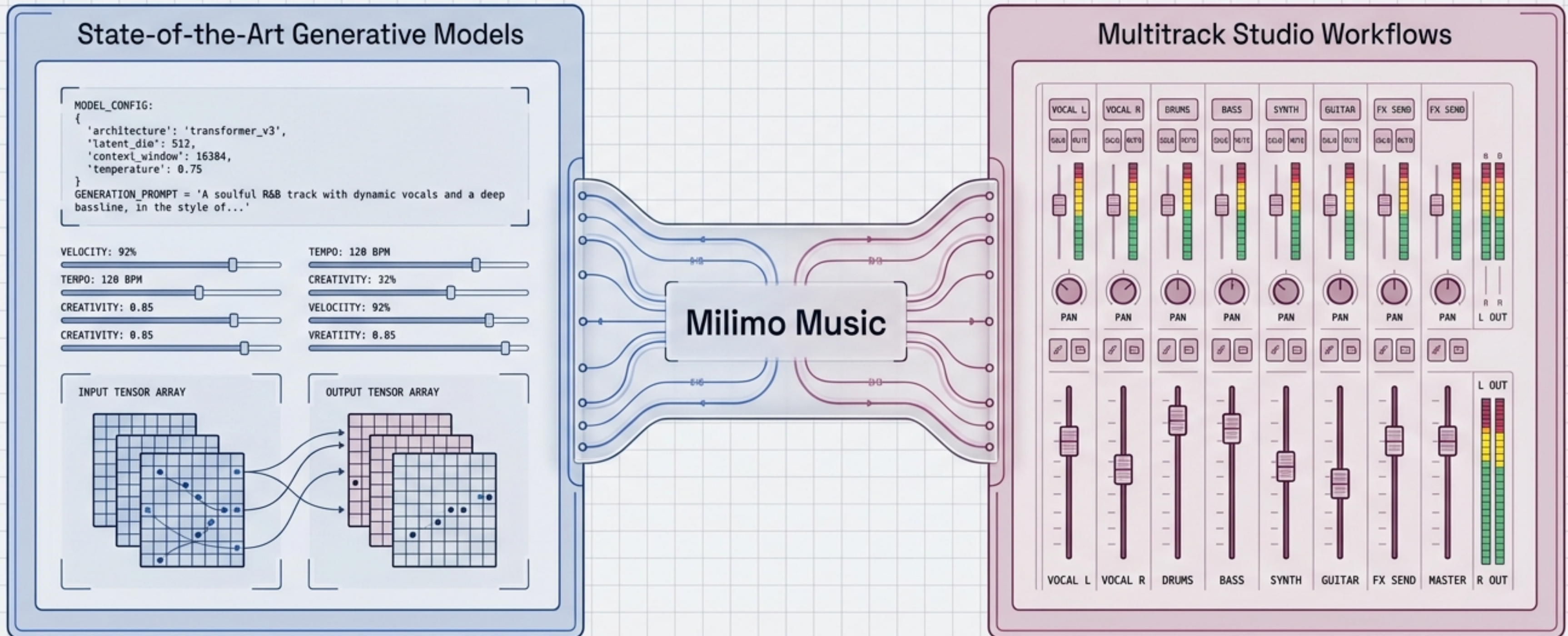
Milimo Music

Next-Generation AI Music Generation, Neural
Transcription & Multitrack Production DAW

Created by Mainza Kangombe

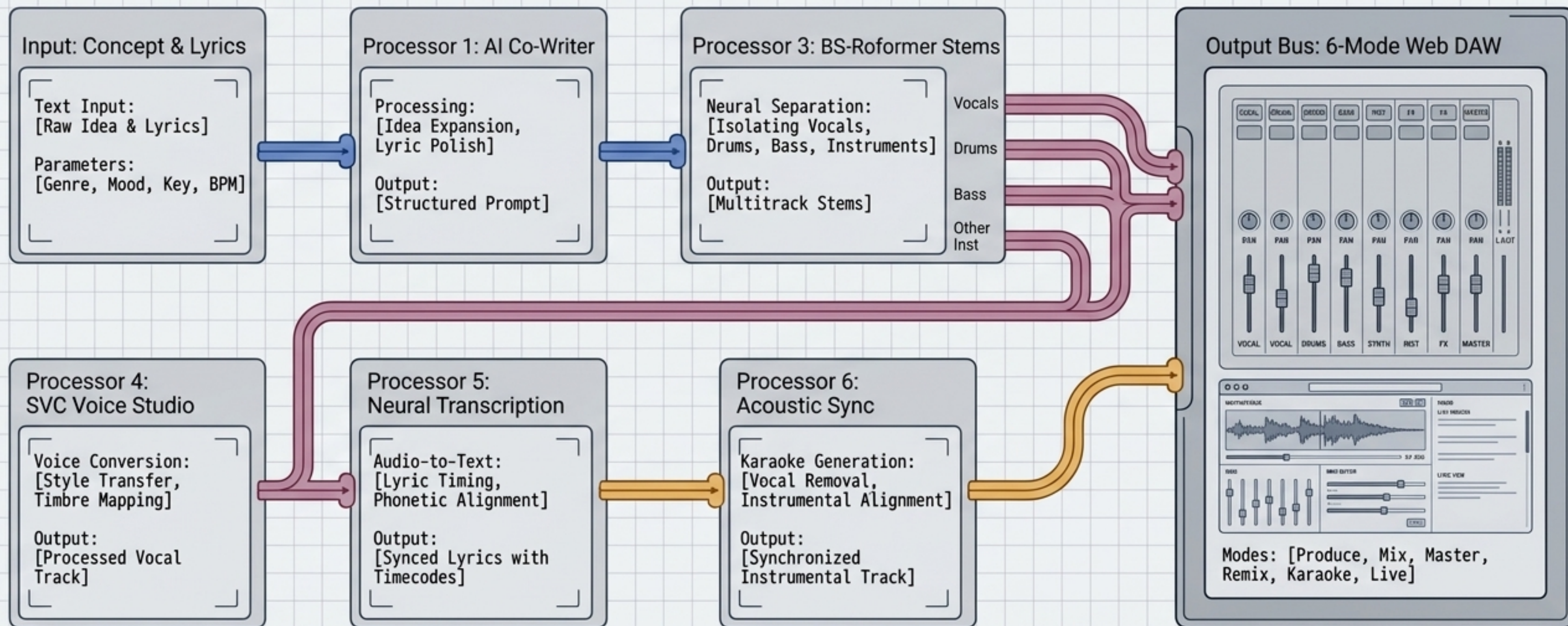
Bridging Raw Inference and Studio Production

Milimo bridges the gap between raw AI model inference and usable, professional DAW environments. It transforms natural language prompts and structured lyrics into mastered stereo tracks, note-level polyphonic MIDI, isolated neural stems, and synchronized acoustic karaoke without leaving the workspace.



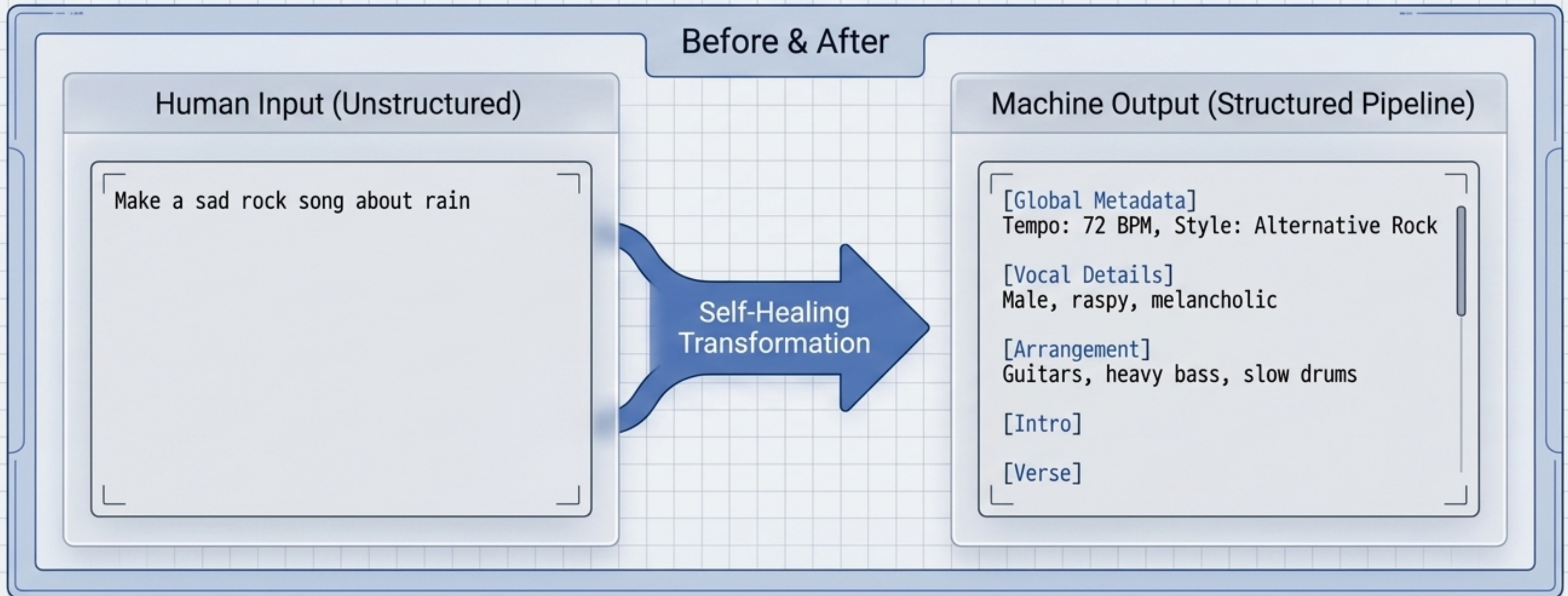
The Signal Flow: From Concept to Master Bus

Modern audio production requires a rigorous, linear pipeline. Milimo routes raw creative intent through seven distinct neural processing stages before culminating in a unified web audio workspace.



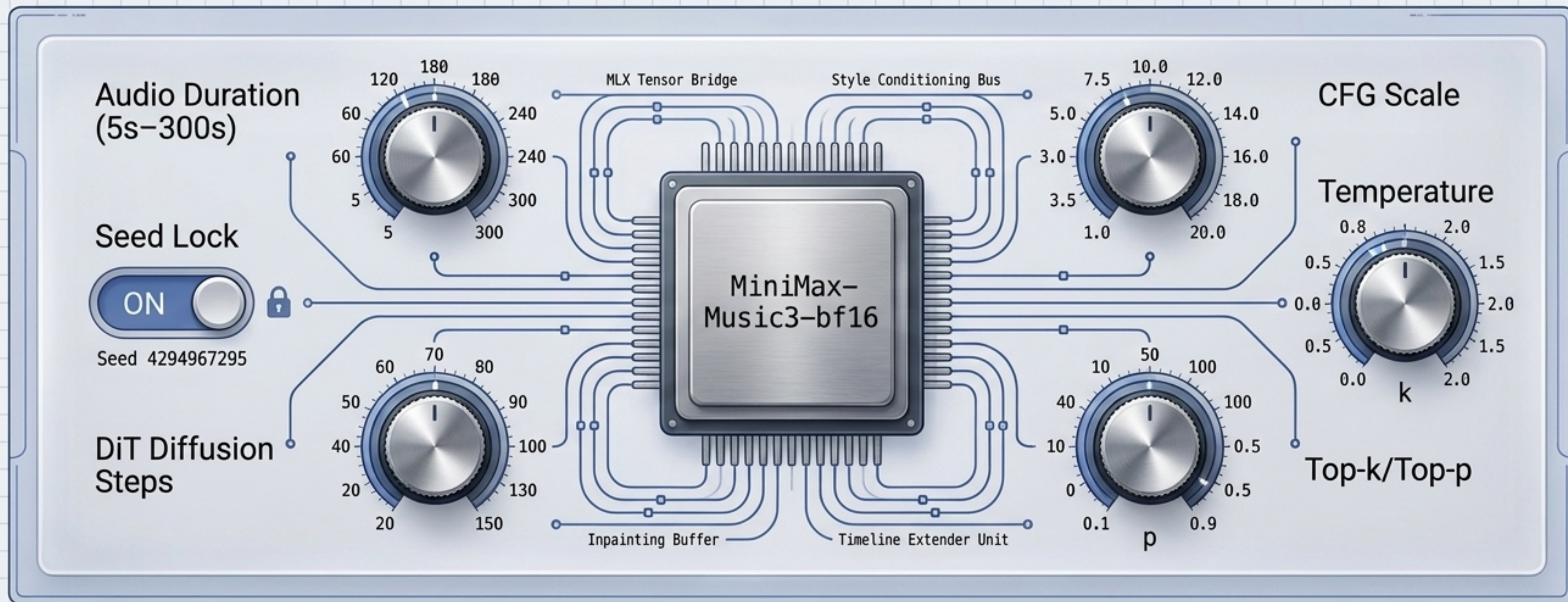
Stage 1: The Self-Healing Producer and AI Co-Writer

The Multi-Provider LLM integration features dynamic model discovery (NVIDIA NIM, DeepSeek, OpenAI, Gemini, OMLX, Ollama). It automatically expands minimalist prompts into professional, structured captions and keyword-aware style recovery to reliably condition the generative engine.



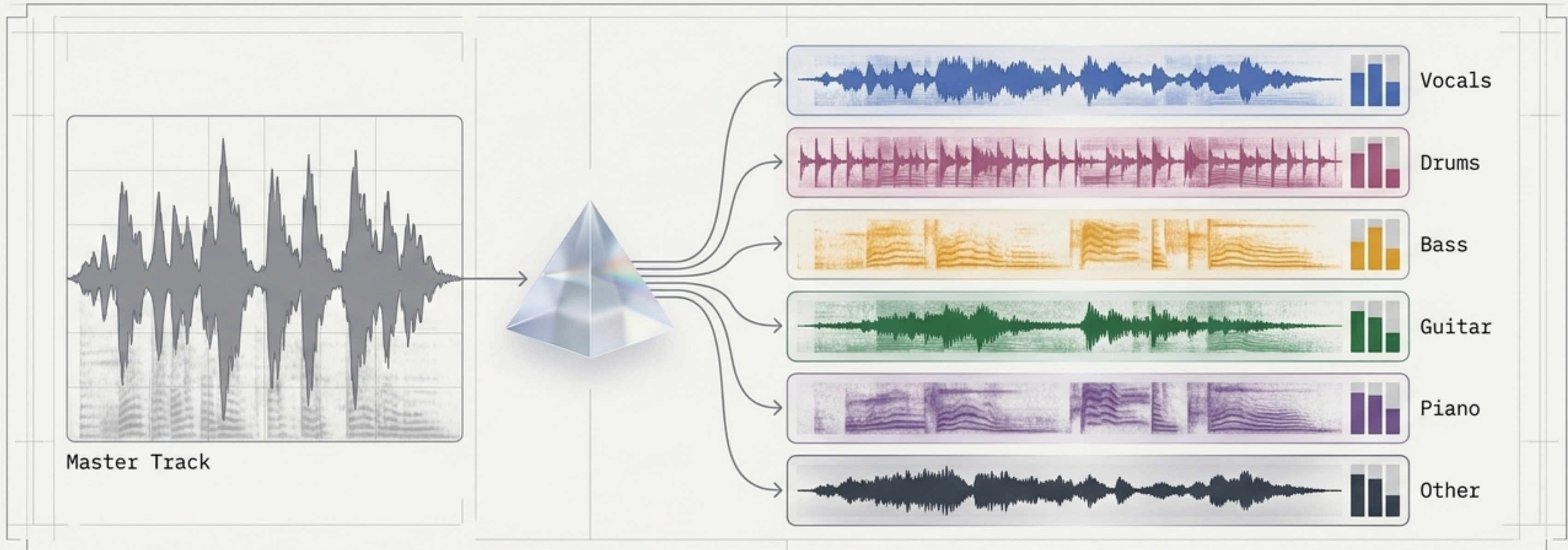
Stage 2: Native Neural Audio Generation

Conditioned on structured multi-section captions and acoustic style descriptors, the core generative engine runs native Apple Silicon MLX weight inference. Features seamless acoustic inpainting and timeline extension to regenerate designated glitch regions or extend track tails.



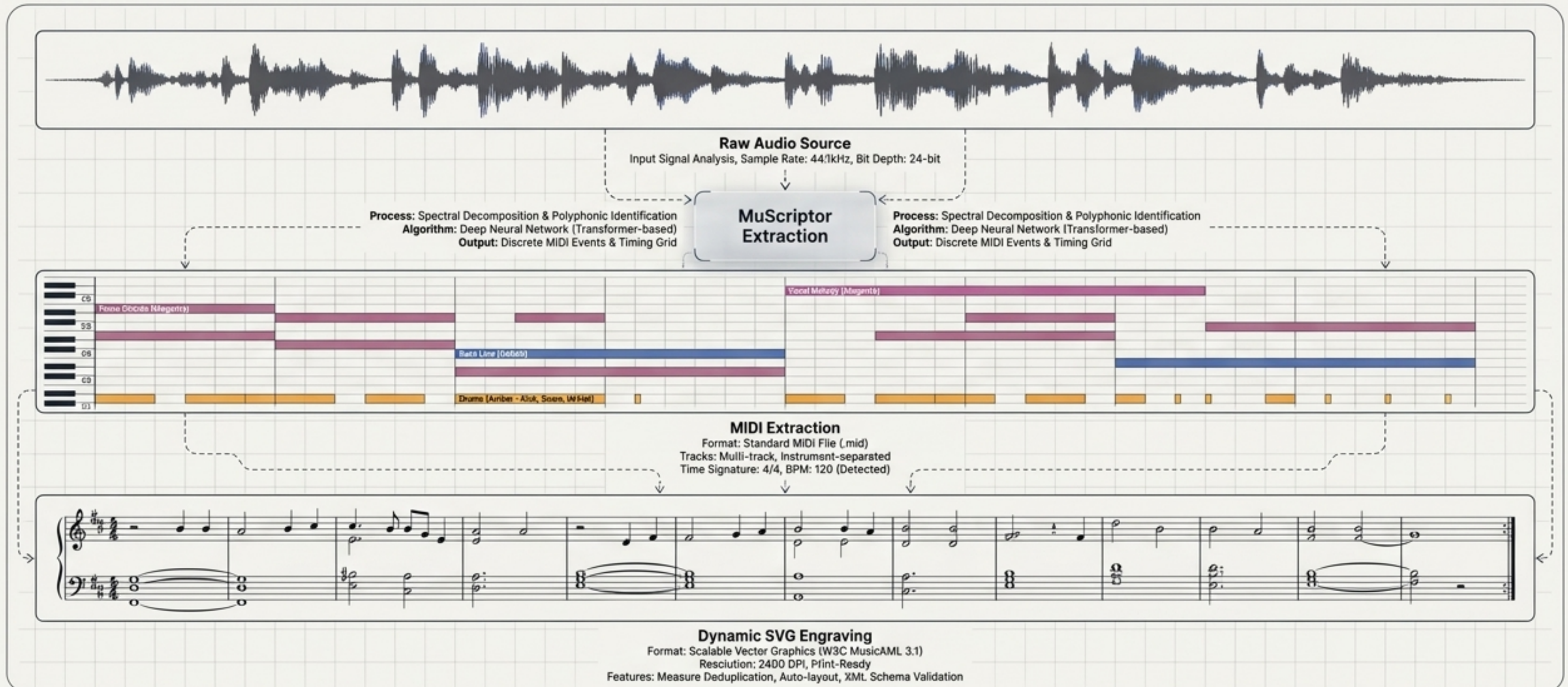
Stage 3: Neural 6-Stem Source Separation

Powered by BS-Roformer and MelBand-Roformer. Automatic native hardware acceleration across CUDA, Apple MPS, and CPU. The Dual-Engine DAW allows toggling between these separated neural stems and dynamic, per-instrument MIDI parts.



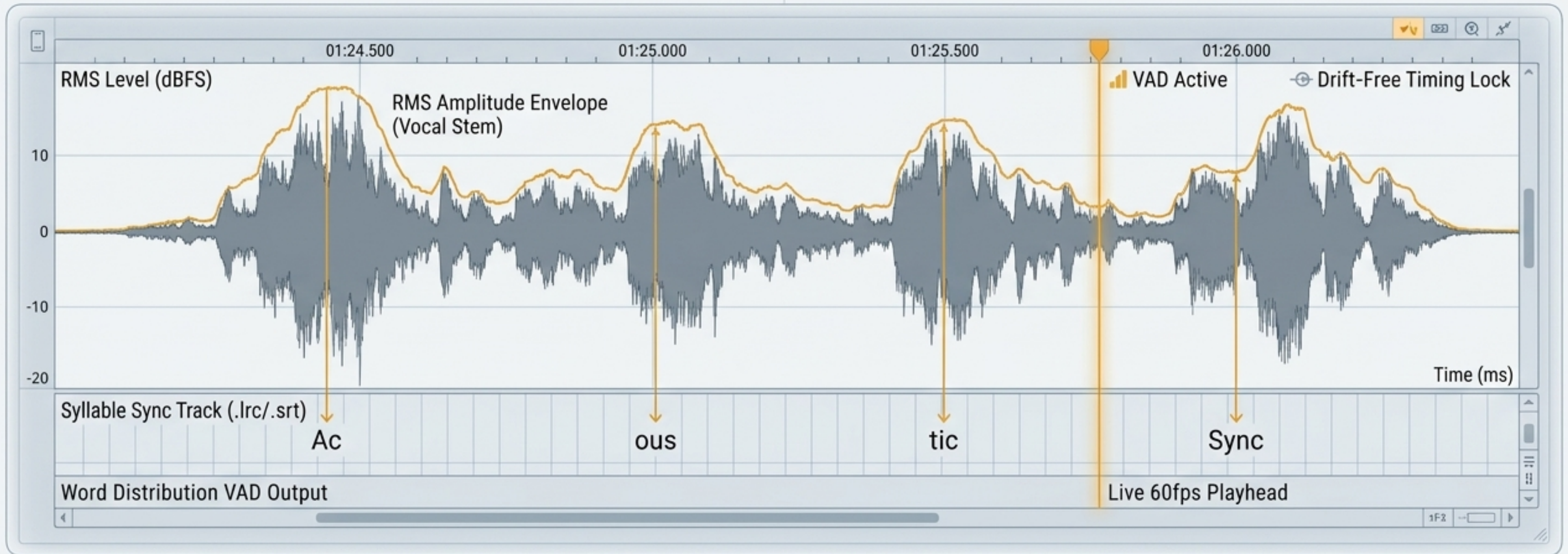
Stage 4: Polyphonic Transcription and Engraving

Polyphonic multi-instrument extraction (Piano, Bass, Drums, Vocal melody). Extracts downbeat grids and BPM via beat-this. Exports multitrack .mid files ready for Logic/Ableton, and dynamic SVG Grand Staff engraving with measure deduplication.



Stage 5: Acoustic Sync and Vocal Energy Extraction

Isolated vocal stems are analyzed for RMS amplitude envelopes, enabling drift-free timing even during long instrumental solos. Voice Activity Detection (VAD) maps syllable-weighted word distributions to a live 60fps playhead, outputting industry-standard .lrc and .srt subtitles.



Stage 6: Consent-Gated Singing Voice Conversion

The SVC Studio allows offline vocal cloning to transform vocal tracks into custom timbres. Crucially, the system utilizes Consent-Enforced Governance via cryptographic audio consent gating to ensure ethical vocal profile creation and prevent unauthorized replication.



The Master Bus: A 6-Mode Web Audio Workspace

The neural pipeline converges in the interactive DAW.

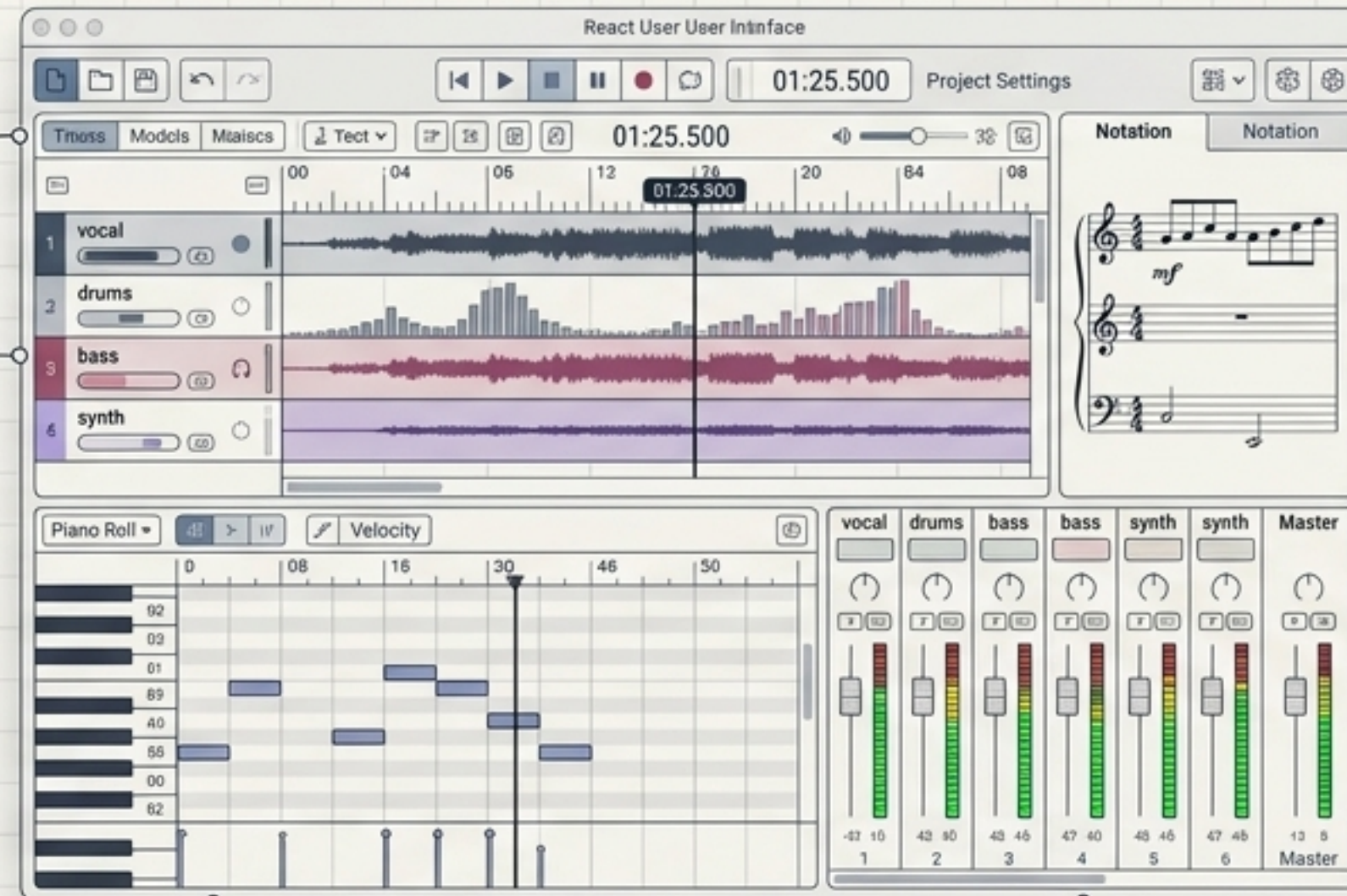
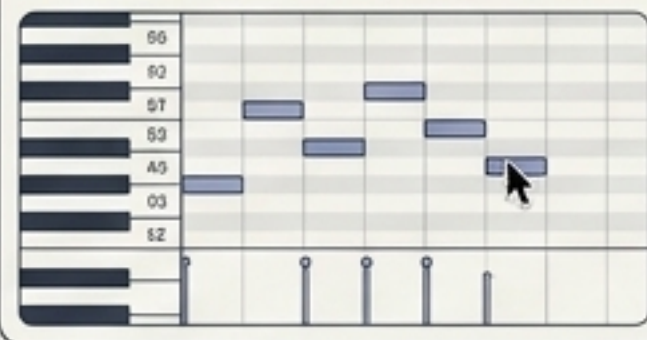
Listen: Live metadata and waveform visualization.



Arrange: Stem lanes with real note-density and measure grids.



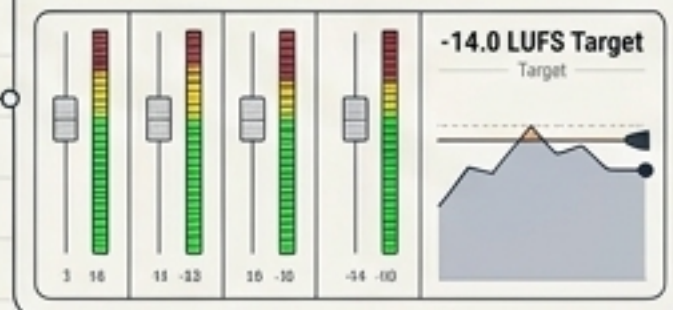
Piano Roll: 144px Apple Studio Grand Keyboard with interactive MIDI.



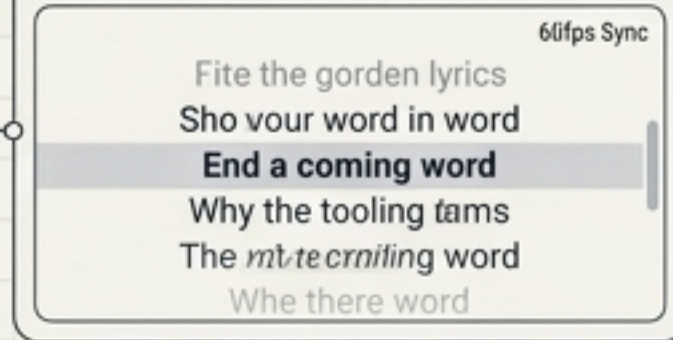
Notation: Dynamic SVG Grand Staff with PDF export.



Mix: Animated LED peak meters and Matching reference target (-14.0 LUFS).

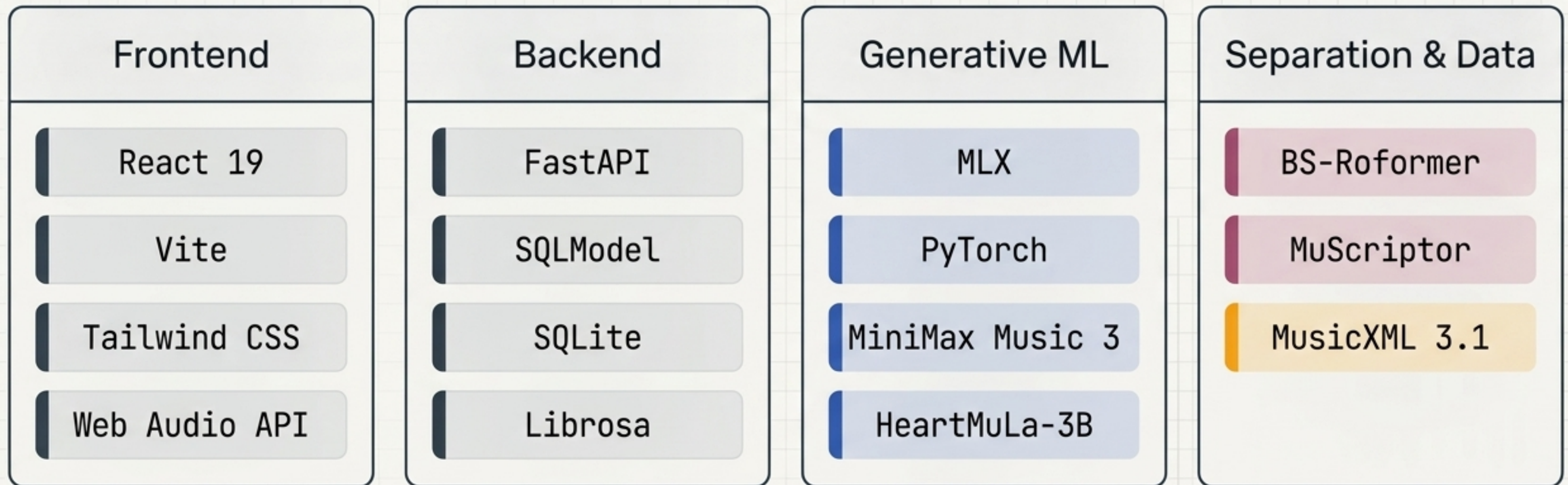


Lyrics: Fullscreen 60fps karaoke teleprompter.



System Architecture and Technology Stack

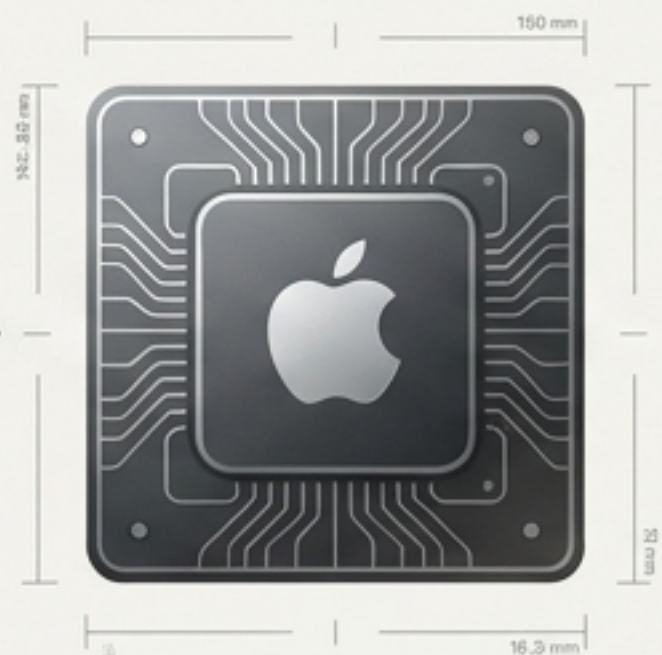
A modular, decoupled architecture ensures high-fidelity frontend audio playback via Web Audio API, powered by a robust Python/FastAPI backend handling asynchronous task execution, SSE progress streaming, and heavy ML inference.



Hardware Acceleration and Asymmetric Deployment

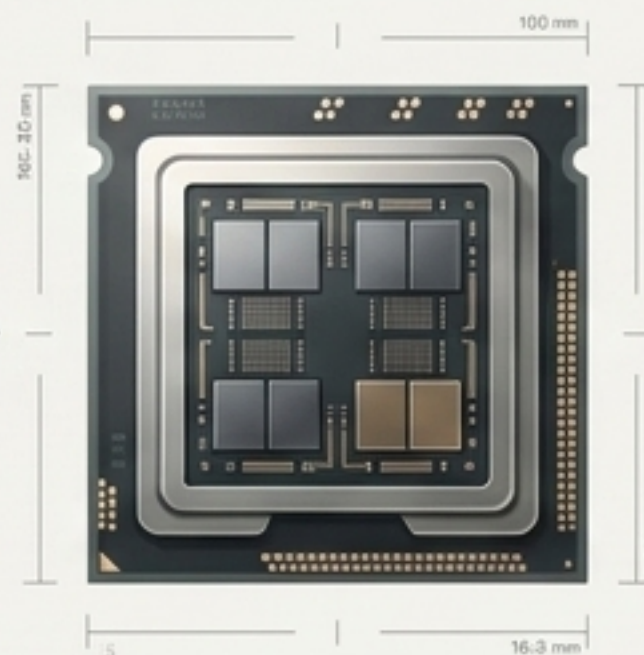
Milimo adapts to host hardware. Apple Silicon (M1-M4) supports native MLX weight inference for full audio generation. On Windows/Linux (CUDA), the DAW, transcription, stems, and agents function fully—but generation utilizes a labeled placeholder synthesizer unless MLX is present.

Apple Silicon Ecosystem



Native MLX Generation + Full Studio Workflow Supported

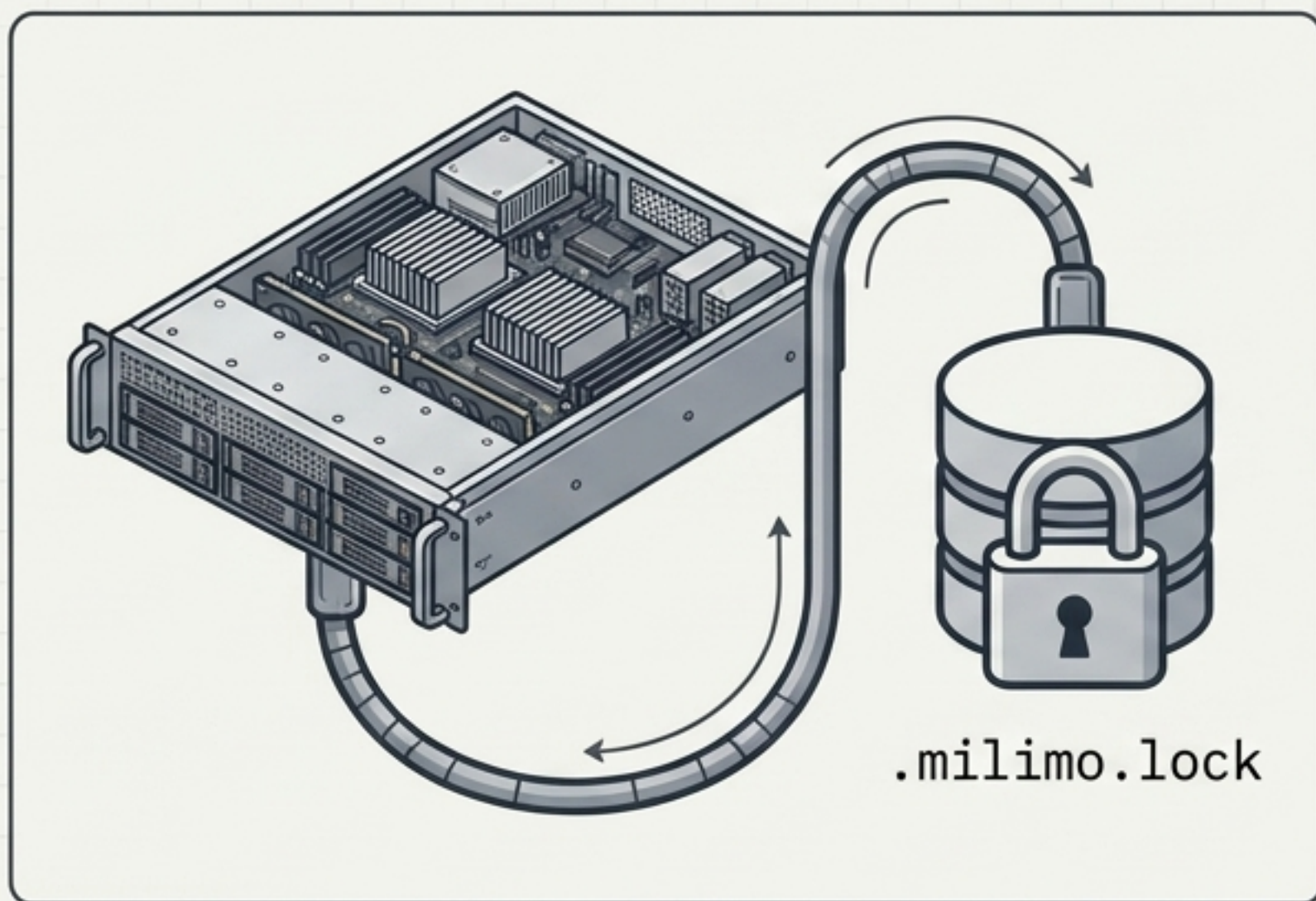
CUDA / Windows / Linux



Full Studio + Stems (Generation falls back to placeholder synth)

Instance Locking and Database Integrity

The backend enforces a strict single-instance lock to prevent GPU and database contention, reclaiming stale locks after 30s. Because SQLite runs in WAL mode, live copying is prohibited; backups require a consistent snapshot.



DANGER:

`MILIMO_ALLOW_MULTI_INSTANCE=1`

```
VACUUM INTO 'backup.db';
```


Component-Level Licensing Governance

Milimo Music embraces open-source while strictly adhering to upstream governance. While the core platform is Apache 2.0, the neural transcription and generative audio weights operate under specific non-commercial and open-weights constraints.



Platform Code

Apache License 2.0



MuScriptor Weights

CC BY-NC 4.0 (Creative Commons Attribution-NonCommercial)



MiniMax Weights

MiniMax Open Weights License

Initializing the Milimo Studio Environment

Prerequisites: Python 3.10+ and Node.js $\geq$ 20.19.

1. Clone

```
git clone --recurse-submodules  
https://github.com/mainza-ai/milimomusic.git
```

Launch Studio Workspace

2. Backend Initialization

```
uvicorn app.main:app --host 0.0.0.0 --port 8000
```

Access API at port 8000

3. Frontend Initialization

```
npm install && npm run dev
```

Access DAW at port 5173

Launch Studio Workspace localhost:5173